



How We Tested

Our test bed was an Intel D875PBZ motherboard running an Intel 3.2 GHz processor. This could test the RAM modules at all clock speeds, from 266 to 400 MHz. A 40 GB Seagate Barracuda hard drive, a GeForce4 TI 4600 and the SoundBlaster Audigy2 were the other components.

We tested only Double Data Rate Synchronous Dynamic Random Access Memory (DDR SDRAM), since all motherboards available now support it, irrespective of clock frequency. Older SDRAM, though still available in the market, is slowly being phased out. The same goes for the RAMBUS technology that Intel banded about for some time before going the DDR way. We carried out five tests—three real-world tests, and two

synthetic ones.

We used SiSoft Sandra, PC Mark 2004, Adobe Photoshop, Content Creation and UT 2003 for our tests. SiSoft Sandra is a synthetic test that emulates read and write on the RAM, and gives integer and floating point scores. It doesn't actually transfer any data on or off the RAM. PCMark 2004, however, does just that, and gives a single, unified score.

The Content Creation test is a system-level benchmark, and was used to obtain scores for configurations of the same system with the RAM being the only variable. The UT 2003 benchmark was used to gauge the gaming performance difference amongst the different data rates of the RAM, in the vertical category.

The value for money offered by Hynix is unmatched by the other manufacturers—it's the lowest priced module in the lot, and gives good performance, making it our value winner. Hynix also happens to be the most widely available in the market.

In the vertical tests we conducted on the Hynix modules, the performance plotted against increasing clock speeds gave a linear graph. The percentage increase in performance, in moving from DDR 266 to DDR 400, on our *Unreal Tournament 2003* tests, was approximately eight per cent. On overall scores, the difference was almost 21 per cent. The price difference is about Rs 1,000, which works out to approximately 33 per cent more.

Conclusion

A RAM module can run at its full clock speed only if the motherboard supports it. If you purchase a DDR 400 RAM stick, and you have a motherboard that only supports DDR 266, the RAM will run at DDR 266. As with processors, when you purchase RAM, you need to consider what you use your computer for. Do you use it primarily for office work with a little gam-

ing thrown in? Or are you the user who dabbles with audio-visual stuffs a lot of gaming?

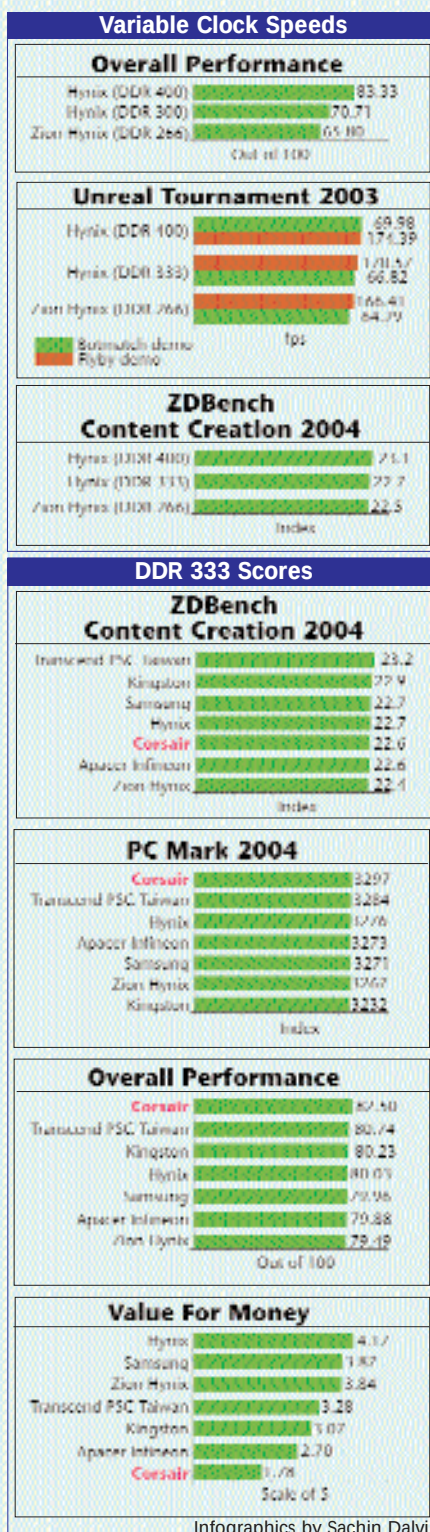
Irrespective of what you do, what remains clear is that with the increasing amount of resources that newer applications demand, RAM plays a critical role. We do not recommend DDR 266, as the price difference compared to DDR 333 is only a few hundred rupees. The DDR 266 module we tested, which had Hynix chips, was from Zion.

For running a mix of applications consisting of office applications, games and movies, the Transcend DDR 333 is highly recommended. If price is a constraint, you might want to look at the Hynix DDR 333.

If you do lots of video and audio editing, work on databases, or if you're the avid gamer who needs the utmost performance out of the computer, DDR 400 is what you will need. Though you'll have to pay a premium, the performance will be worth it—of course, provided you have a system that will employ the RAM at full steam. ■

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